
AoI and SW-UCB diagnosis: 48,000 existing slots, seed 0
(a) Cumulative averages include all failed transmissions
SW-UCB
Random
Fixed (a=4)
Cumulative weighted-sum AoI (slots)
Time (s)
(b) Per-slot AoI: infrequent, large spikes drive the average gap
Per-slot weighted-sum AoI (slots, log scale)
Time (s)
(c) At a=10, cohort 1 is muted during the 93.10-93.88 s outage
SW-UCB: per-slot AoI
157.81 -> 171.10
Peak = 3297.49
Fixed (a=4): per-slot AoI
79 consecutive power-outage slots
SW-UCB: cumulative average
Per-slot weighted-sum AoI (slots)
Cumulative average AoI (slots)
Time (s)